Richard Álvarez

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Education

Kenyon College Aug 2020 – May 2024
Bachelor of Arts in Film; GPA: 3.4/4.0 *Gambier, Ohio*
Minor in History and Concentration in Integrated Program in Humane Studies

Authored two papers on machine learning applications in creative industries. Produced and edited over 15 experimental video projects, including music videos, audio-reactive visualizations, and short films.

Relevant coursework includes Senior Research Seminar, AI for the Humanities, Advanced Post-Production, Data Structures and Program Design, Digital Photography, Research Strategies in the Contemporary Age, and Software Development.

Web Projects

Joaquin Morales Jun 2025
Portfolio *joaquinmoralesdp.com*
Designed and deployed a dynamic portfolio site for a professional cinematographer. The project used Next.js for high performance, with Tailwind CSS for responsive design. Implemented a custom CMS to enable efficient content updates, managing galleries and testimonials with ease. Leveraged DigitalOcean S3 storage for scalability and fast load times for video and photo content.

Machine Television Oct 2024
Online Store *machinetelevision.com*
Developed a functional e-commerce platform for an independent skate brand. The site was built using Next.js and Tailwind CSS for an intuitive front-end, paired with Node.js for a robust back-end infrastructure. Integrated Stripe API for seamless payment processing, optimizing user workflows across desktop and mobile.

GREasyVocab Flashcards Jul 2024
Web App *Inactive*
Created a personalized GRE vocabulary tool powered by OpenAI's APIs. The application leverages LangChain to provide personalized prompts tailored to user inputted data. Developed a secure full-stack system with user authentication and database management, ensuring a smooth and customized learning experience.

Skills

Programming: Python, JavaScript/TypeScript, SQL, C/C++
Backend & Databases: Node.js, MySQL, MongoDB
Frameworks: Next.js, React, Tailwind CSS, Scikit-Learn, Keras, pandas, NumPy, LangChain
Machine Learning: Diffusion models, LoRA, LxMs, retrieval-augmented generation (RAG), fine-tuning LLMs
Visualization: DaVinci Resolve, Tableau, Adobe Creative Suite, Blender

Publications

A Retrieval-Augmented Film Recommendation System *GitHub*
Digital Kenyon *May 8th 2024*
This project utilized LangChain’s OpenAI integration to dynamically generate queries based on user preferences, showcasing the potential of advanced AI and machine learning in digital entertainment. The Retrieval-Augmented Film Recommendation System was developed using Node.js and integrated with the OMDb and TMDb APIs to enhance movie metadata, delivering precise and personalized recommendations.

Captivating Video Editing Secrets *GitHub*
Digital Kenyon *May 23rd 2023*
Developed an analysis of short-format video editing trends by employing PySceneDetect for feature extraction and Deep Neural Networks (DNNs) for unsupervised clustering of videos, yielding insights into innovative editing techniques and creative decisions.